

CHAPTER

3

Printing

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LEARNING OBJECTIVES

- **To understand the styles of printing**
- **To get an insight into different printing methods**
- **To be familiar with the after treatment processes for printed fabrics**

3.1 Introduction

The art of printing fabric was known as early as 300 BC (BCE) Printing is the art of colouring the surface of any item. Tattooing of body is one of the most common printing of olden days. The impression of object dipped in dyes on fabric is the basic technique of printing. Textile printing is defined as the 'localized dyeing' or restricted form of dyeing a particular area of cloth or design. Dyes or pigments are applied to produce attractive patterns or designs with one or more colours. Printing is quicker and cheaper method of colouring fabrics. Generally a pigment or paste is needed to print textiles. Printing is carried by different methods namely block, screen, stencil etc.

3.2 Printing Paste

In printing, dyes or pigments are applied in the gel form to prevent the flowing of print design during printing and subsequent drying. Dyes are thickened by mixing it with gums or starches. This thickened dye solution is called as print paste. Print paste is composed of dyestuff, thickener, hygroscopic agents and auxiliary chemicals. Thickeners are added to improve the viscosity and better penetration of the dyestuff into the fabric. The thickener used for print paste preparation may be natural like starch, gum Arabic or synthetic polymers like polyvinyl alcohol and polyacrylamide. Hygroscopic agents used for print paste preparation are water soluble substances like urea and glycerine. They help the dye to enter into the fibre structure for fixation. Auxilliary chemicals such as solvents improve dye solubility and colour yield. Additional chemicals may be added depending on the fibres and dyes. For example, citric acid may be added for acid dyes or alkali added for reactive dyes. Thickness and freshness of the printing paste are two important aspects to be considered for the quality and durability of printing.

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